

GRUPO 6:

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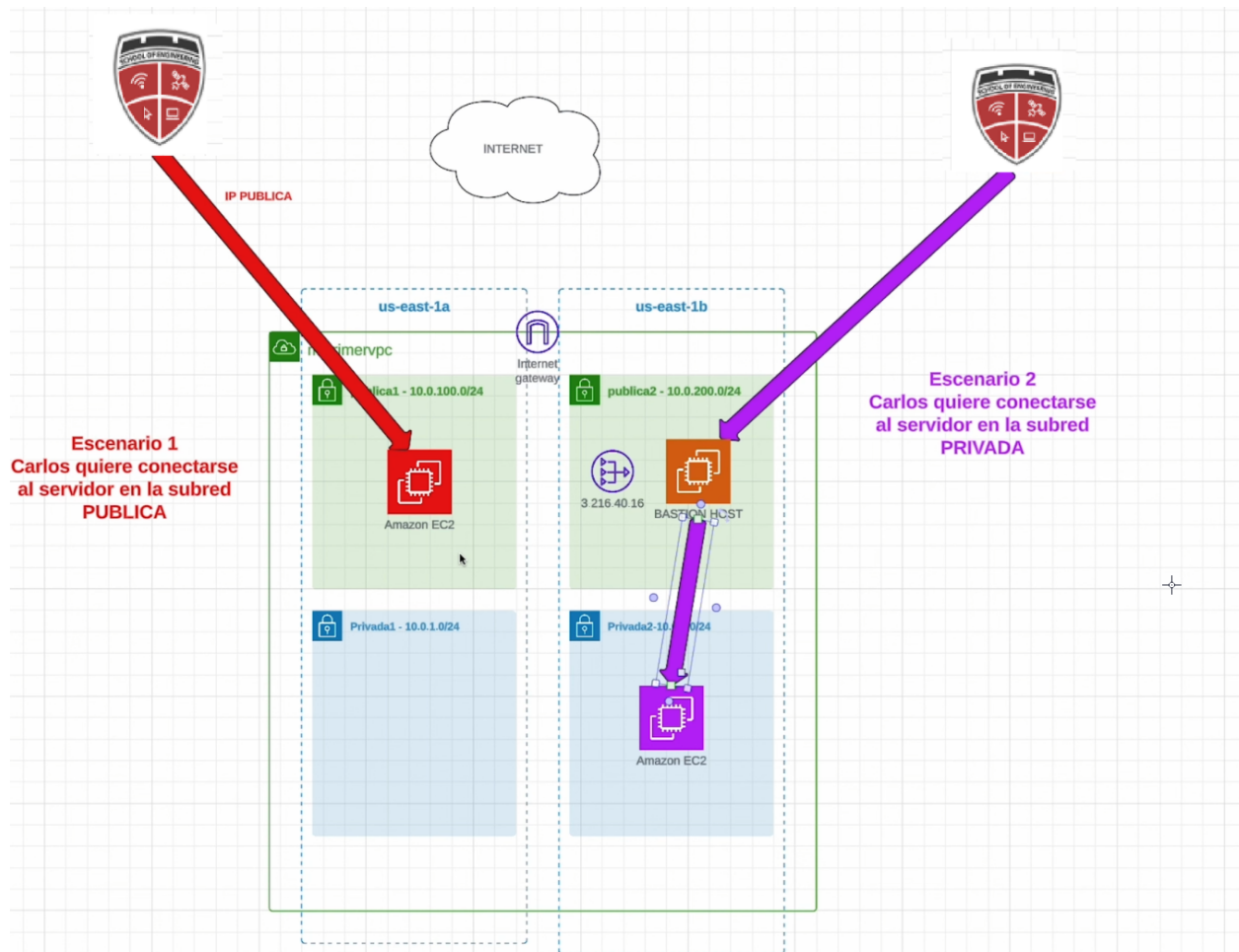
Jose Antonio Mendoza Aquino

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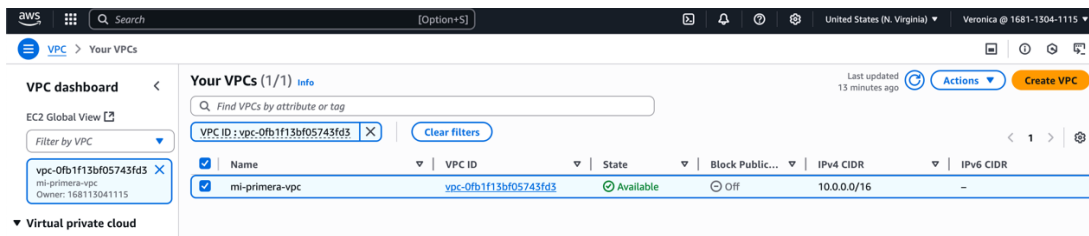
CREAR UNA ARQUITECTURA DE RED EN AWS CON IAM y EC2

1. ESQUEMA INICIAL CREAR LAS SUBREDES Y LOS ENRUTAMIENTOS DEACUERDO A LO SIGUIENTE:



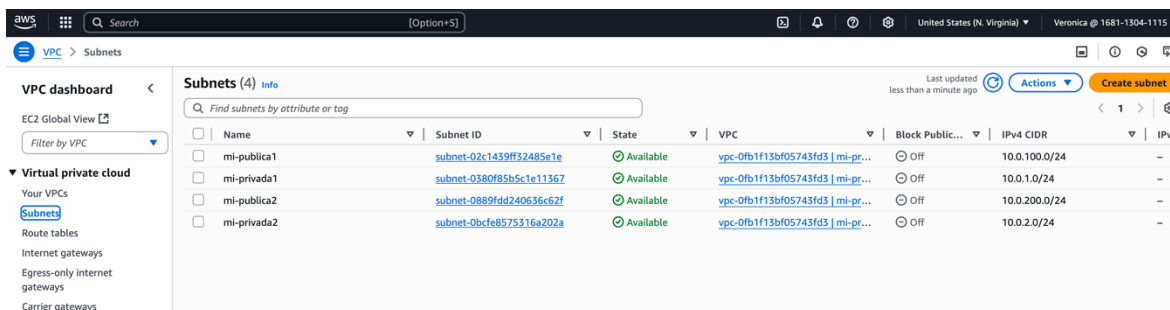
SOLUCIÓN

1. Creación de VPC

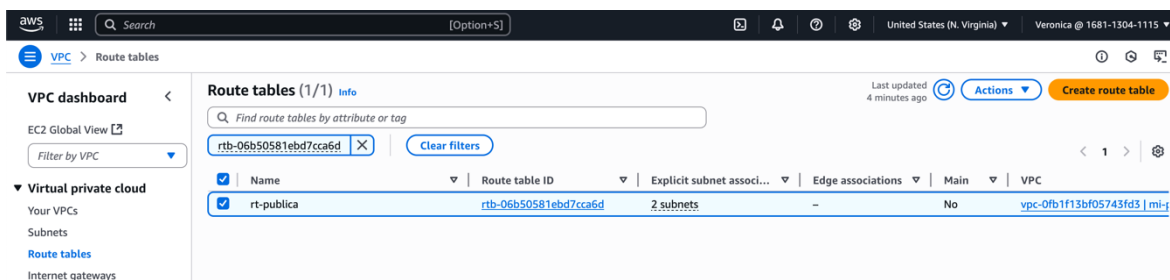


2. Creación de subredes:

- publica1
- privada1
- publica2
- privada2



3. Creación de tabla de enrutamiento para la red pública



4. Creación de Internet Gateway

Internet gateways (1) Info

Find internet gateways by attribute or tag

VPC ID: vpc-0fb1f13bf05743fd3

Name	Internet gateway ID	State	VPC ID	Owner
mi-igw-mivpc	igw-0bf35901c11122a90	Attached	vpc-0fb1f13bf05743fd3 mi-primera-vpc	168113041115

5. Resultado VPC con subredes y tabla de enrutamiento

Resource map Info

VPC Show details
Your AWS virtual network

mi-primera-vpc

Subnets (4)
Subnets within this VPC

- us-east-1a
 - mi-publica1
 - mi-privada1
- us-east-1b
 - mi-publica2
 - mi-privada2

Route tables (2)
Route network traffic to resources

- rt-publica
 - 2 subnet associations
 - 2 routes including local
- rt-privada

Network connections (1)
Connections to other networks

- mi-igw-mivpc

6. Creación de grupos de seguridad

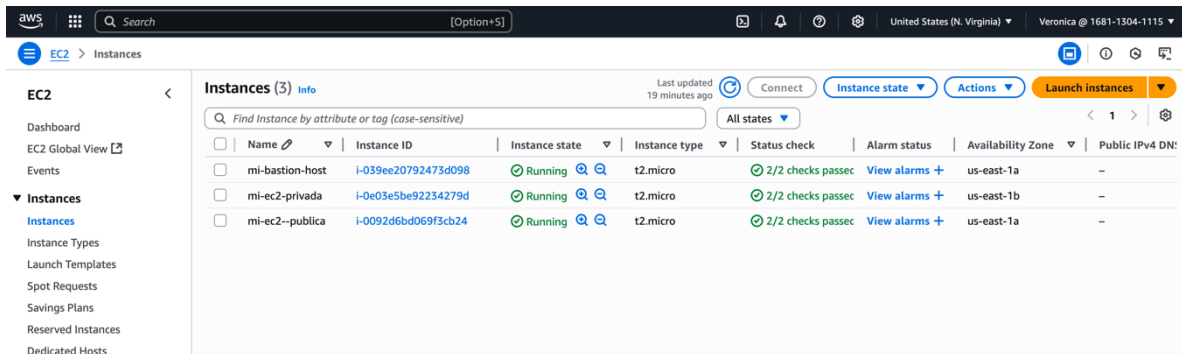
Security Groups (3) Info

Find security groups by attribute or tag

mi

Name	Security group ID	Security group name	VPC ID	Description
mi-sg-bastion	sg-07d9e4d743f552b0f	mi-sg-bastion	vpc-0fb1f13bf05743fd3	sg-bastion
mi-sg-privada	sg-043e82975faf94fe0	mi-sg-privada	vpc-0fb1f13bf05743fd3	mi-sg-privada
mi-sg-publico	sg-0521d952c6fa3475a	mi-sg-publico	vpc-0fb1f13bf05743fd3	mi-sg-publico

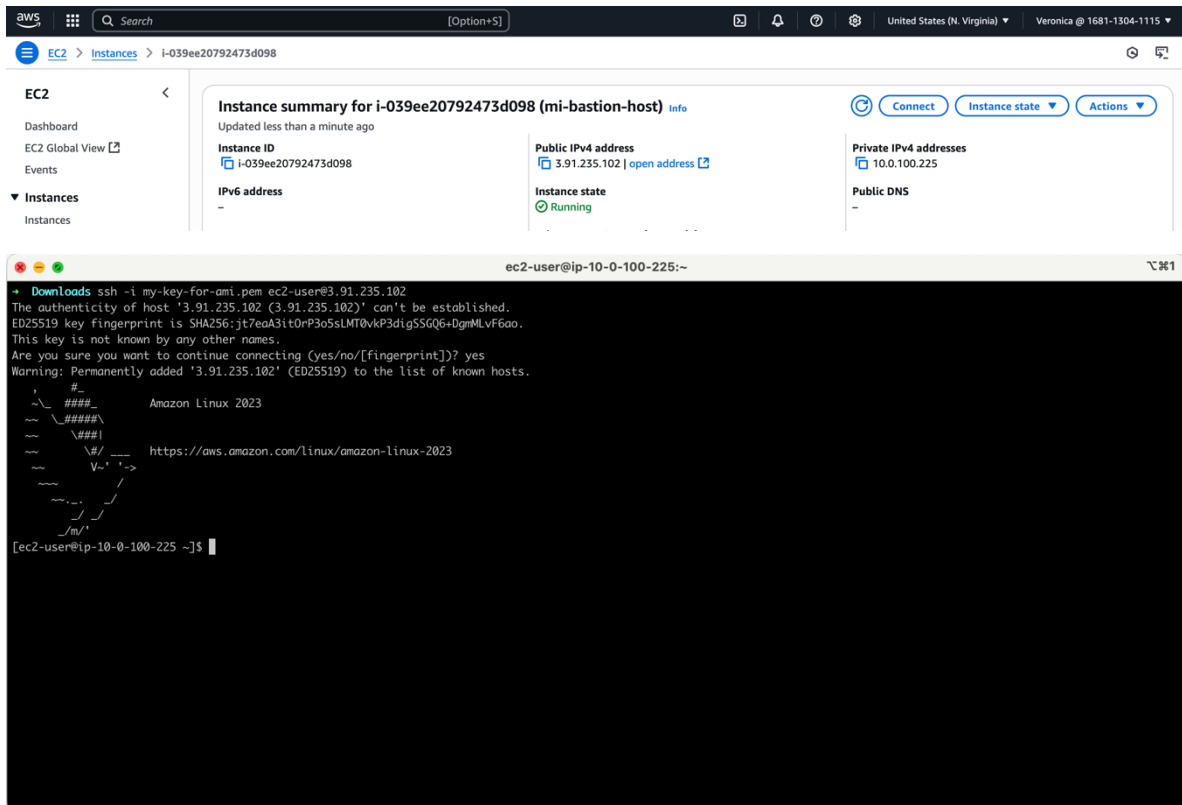
7. Creacion de instancias EC2



The screenshot shows the AWS Management Console for the 'Instances' page. The left sidebar contains navigation links for EC2, Dashboard, EC2 Global View, Events, and a list of instance types (t2.micro, t2.xlarge, etc.). The main content area displays a table of instances with columns for Name, Instance ID, Instance state, Instance type, Status check, Alarm status, Availability Zone, and Public IPv4 DN. Three instances are listed: 'mi-bastion-host' (ID: i-039ee20792473d098), 'mi-ec2-privada' (ID: i-0e03e5be92234279d), and 'mi-ec2-publica' (ID: i-0092d6bd069f3cb24). All instances are in the 'Running' state.

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4 DN
mi-bastion-host	i-039ee20792473d098	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-
mi-ec2-privada	i-0e03e5be92234279d	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1b	-
mi-ec2-publica	i-0092d6bd069f3cb24	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a	-

8. Conexión a IP pública de la instancia de MI-BASTION-HOST



The screenshot shows the AWS Management Console for the 'Instances' page, specifically the 'Instance summary' for 'mi-bastion-host' (ID: i-039ee20792473d098). The summary displays the instance's public IPv4 address (3.91.235.102), private IPv4 addresses (10.0.100.225), and its running state. Below the console, a terminal window shows an SSH connection to the instance using the command 'ssh -i my-key-for-ami.pem ec2-user@3.91.235.102'. The terminal output shows the SSH client's warning about the host's authenticity and the user's confirmation to proceed. The terminal prompt is '[ec2-user@ip-10-0-100-225 ~]\$'.

```
ec2-user@ip-10-0-100-225:~$ ssh -i my-key-for-ami.pem ec2-user@3.91.235.102
Warning: Permanently added '3.91.235.102' (ED25519) to the list of known hosts.
ec2-user@ip-10-0-100-225:~$
```

9. Conexión a IP publica de la instancia MI-EC2-PUBLICA

The screenshot displays the AWS Management Console interface. On the left sidebar, the navigation menu includes "EC2", "Dashboard", "EC2 Global View", "Events", and "Instances". The main content area shows the "Instance summary for i-0092d6bd069f3cb24 (mi-ec2--publica)". Below the summary, there are three tabs: "Connect", "Instance state", and "Actions". The instance details section lists:

- Instance ID:** i-0092d6bd069f3cb24
- Public IPv4 address:** 54.80.136.254 | open address
- Private IPv4 addresses:** 10.0.100.88
- IPv6 address:**
- Instance state:**
- Public DNS:**

Below the console view, a terminal window titled "ec2-user@ip-10-0-100-88:~" shows the following commands and output:

```
[ec2-user@ip-10-0-100-225 ~]$ exit
logout
Connection to 3.91.235.102 closed.
➔ Downloads ssh -i my-key-for-ami.pem ec2-user@54.80.136.254
The authenticity of host '54.80.136.254 (54.80.136.254)' can't be established.
ED25519 key fingerprint is SHA256:OWk81144SdAFgeWPJ+2KQodxl7cs7BTT047s3il3dvA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '54.80.136.254' (ED25519) to the list of known hosts.
```

The terminal also shows the Amazon Linux 2023 logo and the URL https://aws.amazon.com/linux/amazon-linux-2023.

10. Conexión a la instancia de red privada MI-EC2-PRIVADA desde MI-BASTION-HOST.

10.1. No es posible conexión directa a la red privada

EC2

Dashboard

EC2 Global View

Events

Instances

Instance summary for i-0e03e5be92234279d (mi-ec2-privada) Info

Connect

Instance state

Actions

Updated 3 minutes ago

Instance ID

i-0e03e5be92234279d

Public IPv4 address

13.220.143.220 | open address

Private IPv4 addresses

10.0.200.29

IPv6 address

Instance state

Public DNS

vtancara@VERONICAs-MacBook-Pro:~/Downloads

Downloads ssh -i my-key-for-ami.pem ec2-user@13.220.143.220

ssh: connect to host 13.220.143.220 port 22: Operation timed out

Downloads

Downloads

Downloads

Downloads

Downloads

10.2. Se copia la llave al servidor del bastion mediante scp

```
vtancara@VERONICAs-MacBook-Pro:~/Downloads
```

```
+ Downloads ssh -i my-key-for-ami.pem ec2-user@3.91.235.102
```

```
_#_
~\   ###_      Amazon Linux 2023
~~~ \#####\
~~~~ \###|
~~~~ \#/  --- https://aws.amazon.com/linux/amazon-linux-2023
~~~~ V~' '->
~~~~ _--_ /
~~~~ _--_ /
~~~~ _m/'
```

```
Last login: Sun Jun 22 14:19:30 2025 from 181.115.172.58
[ec2-user@ip-10-0-100-225 ~]$ exit
logout
Connection to 3.91.235.102 closed.
```

```
+ Downloads pwd
/Users/vtancara/Downloads
```

```
+ Downloads scp -i my-key-for-ami.pem mi-key.pem ec2-user@3.91.235.102:/home
scp: stat local "mi-key.pem": No such file or directory
```

```
+ Downloads scp -i my-key-for-ami.pem mi-key-for-ami.pem ec2-user@3.91.235.102:/home
scp: stat local "mi-key-for-ami.pem": No such file or directory
```

```
+ Downloads scp -i my-key-for-ami.pem my-key-for-ami.pem ec2-user@3.91.235.102:/home
scp: dest open "/home/my-key-for-ami.pem": Permission denied
scp: failed to upload file my-key-for-ami.pem to /home
```

```
+ Downloads scp -i my-key-for-ami.pem my-key-for-ami.pem ec2-user@3.91.235.102:
my-key-for-ami.pem                                100% 1674    12.8KB/s   00:00
```

```
+ Downloads █
```

10.3. Conexión a la instancia de red privada MI-EC2-PRIVADA desde el MI-BASTION-HOST.

The screenshot displays the AWS Management Console interface for an EC2 instance. The top navigation bar shows the 'Instances' page for the instance ID i-0e03e5be92234279d. The instance summary section shows the instance is in the 'Running' state, updated 1 minute ago. It lists the instance ID, public IPv4 address (13.220.143.220), and private IPv4 address (10.0.200.29). Below the summary, a terminal window shows the user logging into the instance via SSH, displaying the Amazon Linux 2023 logo and the URL https://aws.amazon.com/linux/amazon-linux-2023.